
AGENTIC ECONOMIC MODELING

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ABSTRACT

We introduce Agentic Economic Modeling (AEM), a framework that aligns synthetic LLM choices with small-sample human evidence for reliable econometric inference. AEM first generates task-conditioned synthetic choices via LLMs, then learns a bias-correction mapping from task features and raw LLM choices to human-aligned choices, upon which standard econometric estimators perform inference to recover demand elasticities and treatment effects. We validate AEM in two experiments. In a large scale conjoint study with millions of observations, using only 10% of the original data to fit the correction model lowers the error of the demand-parameter estimates, while uncorrected LLM choices even increase the errors. In a regional field experiment, a mixture model calibrated on 10% of geographic regions estimates an out-of-domain treatment effect of -65 ± 10 bps, closely matching the full human experiment (-60 ± 8 bps). Under time-wise extrapolation, training with only day-one human data yields -24 bps (95% CI: [-26, -22], $p < 1e-5$), improving over the human-only day-one baseline (-17 bps, 95% CI: [-43, +9], $p = 0.2049$). These results demonstrate AEM’s potential to improve RCT efficiency and establish a foundation method for LLM-based counterfactual generation.

Keywords Econometrics · Large Language Model · Causal Inference · Bias Correction

1 Introduction

At its core, making business decisions requires understanding cause and effect. For example, when an e-commerce company considers applying new features, we need to understand counterfactuals: how will new features cause changes in customers’ purchasing decisions? The fundamental challenge is that we can never directly observe both what happens when we make a change and what would have happened if we hadn’t. Traditionally, this required either running extensive experiments, conducting surveys, or applying complicated causal inference techniques to historical data.

Large Language Models (LLMs) offer a promising solution through their ability to generate synthetic responses based on vast amounts of world knowledge. The “homo silicus” paradigm [15] establishes that LLMs can serve as simulated economic agents, while [5] shows that fine-tuned LLMs can predict human behavior across diverse cognitive tasks. Recent work further explores how theory-guided LLM personas can generalize across strategic environments [26]. More broadly, LLMs can take on multiple roles in experimental research—from study design to measurement to simulation [8, 11], and evidence suggests that human–LLM hybrids can accelerate insight generation in marketing science [3]. For economists, this opens an attractive possibility [19, 29]: replacing expensive experiments with large-scale synthetic choice data while still recovering demand elasticities and treatment effects.

However, a fundamental challenge remains: LLM-generated choices often diverge systematically from real customer behavior. Off-the-shelf LLMs exhibit systematic preference bias and lower heterogeneity [10, 12]. Models display

machine bias with low within-topic variance [6], and alignment choices shape results systematically [25, 20]. As a result, naive inference on synthetic LLM outputs without bias correction is unreliable [30, 13, 23].

Our approach builds on the prediction-powered inference (PPI) framework [1, 2], which establishes rigorous methods for combining ML predictions with small labeled datasets while maintaining valid confidence intervals. [17] extend PPI to show that AI predictions can serve as valid surrogates when properly calibrated. [24] provide a foundational econometric framework distinguishing prediction problems (where LLMs excel) from estimation problems (where human calibration is essential).

In this paper, we introduce Agentic Economic Modeling (AEM), that aligns synthetic LLM choices with small-sample human evidence for reliable econometric inference. The key innovation builds upon three ideas: First, we recognize that LLMs, having absorbed vast world knowledge, can help us better predict customer behaviors. Rather than using raw LLM predictions directly, we transform their knowledge-rich but biased outputs into precise estimates of human decision-making, building on recent work that formalizes bias correction through learned mappings between human and LLM responses [30]. Second, drawing on the random coefficient discrete choice literature [27], we recognize that community-level choices are better represented by mixtures of individual agent decisions rather than a single agent model. Mirroring the heterogeneity of real communities, we generate predictions from multiple AI personas, each reflecting a distinct decision-making style. Manning and Horton [26] demonstrate that persona mixtures can capture population-level heterogeneity, while Mixture-of-Personas [7] formalizes this as a probabilistic framework. Related work shows that varying personas can reduce some biases in collective choice tasks, though trade-offs remain between diversity and alignment [21, 22]. Third, we leverage this mixture model insight to correct for LLM biases. Rather than trying to make individual LLMs perfectly mimic human behavior, we generate diverse synthetic agents through different personas, and learn the optimal weights to match observed choice patterns. This allows us to span the space of possible decision rules while anchoring predictions to reality using minimal real-world data.

We validate our framework in two settings. First, in a large-scale conjoint experiment (millions of observations), bias correction significantly improves estimation accuracy using just 10% of human data, extending prior work [30]. Then, in a regional field experiment, our mixture model estimated a treatment effect of -65 bps (± 10 bps) in out-of-domain regions, closely matching the actual human experiment effect (-60 ± 8 bps). These results demonstrate AEM’s potential to improve RCT efficiency and establish a foundation method for LLM-based counterfactual generation.

Our strongest results come from this regional setting, where the key challenge is the extensive scope typically required: 10% of regions in treatment for 8+ weeks, with substantial operational impact. Our framework offered a powerful alternative: using stratified sampling, we randomly selected just 10% of ZIP3s in the U.S. for calibration and tested whether our synthetic data could predict behavior in the remaining regions. The mixture model significantly outperformed simpler approaches. We further tested whether we could reduce RCT duration through time-wise extrapolation. By using only one day of human experimental data for bias correction, we achieved a treatment effect estimate of -24 bps with tight confidence intervals [-26bps, -22bps] and high statistical significance ($p < 1e-5$). This significantly outperformed using one day of human data alone (-17 bps, [-43bps, +9bps], $p=0.2049$). All 95% bootstrap CIs are computed over the entire Generation–Correction–Inference pipeline: in 20 replications we resample personas (and regions for region-wise), retrain the bias-correction model, rerun inference, and form a t-based CI—thus accounting for pipeline-wide variability rather than nominal sampling error alone (Section 5.5). We argue that the precision arises from anchoring a small amount of human evidence and then learning to reweight many low-cost LLM simulations, considering all uncertainty in the pipeline.

These results suggest a new approach to running RCTs more efficiently. While maintaining the fundamental benefits of randomization - our framework dramatically reduces the required scale and duration of experiments. The approach shows promise for even more complex RCT settings, from using bias correction across different types of randomized trials to combining multiple small experiments for broader strategic decisions.

This research advances beyond existing work viewing LLMs primarily as economic theory tools [15] by developing a rigorous framework for bridging synthetic and empirical research. As LLMs continue to evolve, our methodology provides a principled approach for leveraging these powerful tools while maintaining scientific credibility. Think beyond our validated applications in RCTs - any causal inference problem that faces missing counterfactual data could benefit from this approach. Ultimately, our framework attempts to bridge the gap between the low-cost, large-scale synthetic choice data produced by LLMs and the high-fidelity requirements of empirical economic inference.

2 Connection between our idea and the Random Coefficient Discrete Choice Modeling

In economic settings, decisions are made by intelligent agents following optimization principles and/or heuristics. A frequently encountered example is from the discrete choice/random utility model (DC-RUM) [27]. The economic agent

is given a number of alternative goods/options to choose from, and each option comes with a vector of attributes $\mathbf{X}$ that are observable to the agent. The agent is commonly modeled as making a choice based on e.g. a softmax model (essentially the single layer perceptron model):

$$\Pr(i \text{ chooses alternative } j) = \frac{\exp(\pi_i^T X_j)}{\sum_k \exp(\pi_i^T X_k)} \quad (1)$$

Agent i here is characterized by the vector of weights π_i . Given this representation, a large literature is geared towards learning the weight vector π_i by minimizing the distance between the (probabilistic) choices generated by this simple agent model, and observed choices of that agent.

While the above model is adequate towards describing the behavior of a single agent, a large number of studies have shown that to explain the behavior of a community of such agents (e.g. set of consumers in a given zip-code or city), a better representation is provided by the *mixture* of choices by individual agents. that is:

$$\Pr(\text{community chooses alternative } j) = \sum_{i \in \mathcal{I}} \theta_i \frac{\exp(\pi_i^T X_j)}{\sum_k \exp(\pi_i^T X_k)} \quad (2)$$

where θ_i represents the weight given to the choices of agent i , reflecting the fact that some agents perhaps have higher income/wealth or have larger families, and hence are responsible for more of the purchases.

This framework is frequently called the “random coefficient discrete choice” framework, where the main idea is to specify e.g. a discrete support for the distribution of agent types, and apply discrete probability weights to represent the choices of the community as a mixture probability generated by these weights.

Where does the “agentic” framework come into play here? We can think of pre-generating a set of N agents, each of whom have a different decision rule, characterized in the above model as π_i . Each pre-generated agent is going to make a choice that is logically consistent with its parameters, but the aim is to calibrate, or fine-tune the population of agents by learning the weights θ_i that best fit the observed choice probabilities in the real world data.

The bigger idea, however, is that this framework is not limited to the very simple single layer perceptron of McFadden, and allows for a plethora of decision rules or agent “personas” that are consistent with what e.g. a large foundational LLM has learned from its immense input set.

Whatever the underlying “agentic” decision maker is, though, our framework calls for a calibration or fine-tuning stage where we learn the weights θ_i for each of the underlying agent personas:

$$\Pr(\text{community chooses alternative } j) = \sum_{i \in \mathcal{I}} \theta_i \Pr(\text{agent persona } i \text{ chooses } j | X) \quad (3)$$

The idea here is that to model the observed set of choices of a community of customers, a single agent persona’s decision framework will typically not be enough to capture the richness of the decision-rule space. However, if we are able to pre-generate a sufficiently diverse set of agent personas, we will be able to fit observe choices by a re-weighting of these decision makers.

This mixture view unifies both settings: in the conjoint analysis, observed customer-level choices allow persona decisions to be corrected and aggregated prior to inference. In the regional experiment, where only regional shares are available, we explicitly learn aggregation weights θ_i to form region-level outcomes.

3 Overall Framework of Agentic Economic Modeling

This section lays out the general LLM experiment pipeline—**Generation–Correction–Inference**, which leverages LLM-generated data anchored to a small set of real human observations to ensure reliable downstream economic inference.

3.1 Generation–Correction–Inference

At its core, our framework aims to solve a fundamental causal inference challenge: estimating how outcomes (Y) respond to interventions (T) in different contexts (x). Technically, we seek to estimate $E[Y|T, X] = g(T, x)$, which

Stage	Input $\rightarrow$ Output	Description
Generation	task descriptions $x \rightarrow$ simulated choices z	Use LLM to produce large-scale synthetic decision data for diverse personas under the specified experiment environment.
Correction	(x, z) , human labels $y \rightarrow$ corrected choices $\hat{y}$	Learn a bias-correction mapping f so that LLM outputs align with human behavior.
Inference	$(x, \hat{y}) \rightarrow$ economic estimates	Run econometric or causal models to derive demand elasticities, treatment effects, etc.

Table 1: Summary of the Generation–Correction–Inference pipeline.

under unconfoundedness has a causal interpretation as $E[Y(T)|x]$, representing the expected outcome if we were to intervene and set T to a specific value.

Our solution transforms this causal inference problem into quantitative estimates through three stages:

First, in the *Generation* stage, we leverage LLMs to create synthetic responses. Instead of using a single prediction, we generate multiple perspectives through different “personas.” Each persona i provides counterfactual predictions $z_i = l_i(T, x)$, drawing on the LLM’s vast knowledge of human decision-making patterns. Next, in the *Correction* stage, instead of using raw LLM counterfactual predictions $l(T, x) = \{l_i(T, x)\}$ for customer behavior directly, we learn a sophisticated function $f = g(l(T, x), c(T, x))$ that incorporates a learned correction function $c(T, x)$ to transform LLM predictions into precise estimates of human decision-making. Specifically, we use a limited human-labeled dataset y to learn a mapping:

$$f : (x, z) \mapsto \hat{y},$$

thereby adjusting LLM outputs to better reflect actual human behavior. Finally, in the *Inference* stage, we treat $(x, \hat{y})$ exactly as we would a fully observed human data set and apply standard econometric or causal models on the corrected pairs $(x, \hat{y})$ to estimate economic quantities θ , such as demand elasticities or treatment effects. A summary of the pipeline is shown in Table 1.

3.2 Two Use Cases in RCT for Agentic Economic Modeling

To validate both the accuracy and scalability of our pipeline, we examine two use cases of increasing complexity. In a *Conjoint Analysis* with millions of observations from tens of thousands of real customers, we model customer-level choices in response to order-level delivery-attribute changes. The bias correction model is trained only on 10% fully-observed customer-level choices to generate the remaining 90% choices synthetically. We then transfer the pipeline to a *Regional Experiment* on delivery options, where treatments are assigned by ZIP3 and only aggregated regional delivery option shares are observed (900 ZIP3s $\times$ 60 days). Thus, We created 17k+ synthetic customer personas for each region to simulate order placements, and learned mixture weights to aggregate persona-level choices into calibrated regional outcomes. Table 5 summarizes the comparison.

4 Agentic Modeling for Conjoint Analysis

4.1 Problem Formulation and Generation Stage for Conjoint Analysis

A brief introduction to conjoint analysis is in Appendix B. Formally, let $c_{primary} = \{c_1, c_2, \dots, c_M\}$ denote a set of primary customers. For each customer c_i , there is an associated profile vector $d_i \in \mathbb{R}^D$. This could include their age, income, and other related demographic information correlated to purchasing preferences.

Let $\mathcal{K} = \{1, 2, 3, \dots, k\}$ denotes the set of possible options for each task. Each customer c_i completes a sequence of conjoint tasks regarding one product represented by $s_i = [(x_{i1}, y_{i1}), \dots, (x_{ij}, y_{ij}), \dots, (x_{it_i}, y_{it_i})]$, where $x_{ij} \in \mathbb{R}^{qk}$ denotes the concatenated features of k options in the j^{th} question; $y_{ij} \in \mathcal{K}$ denotes the customer’s response; q is the number of attributes of one option in a task and t_i is the total number of tasks completed by c_i . Separately, a LLM is tasked with completing the same set of conjoint tasks, producing choices denoted by $l_i = [(x_{i1}, z_{i1}), \dots, (x_{ij}, z_{ij}), \dots, (x_{it_i}, z_{it_i})]$, where $z_{ij} \in \mathcal{K}$ represents the LLM’s predicted choice for the j^{th} question. The entire primary dataset $\mathcal{D}_{primary} = \{(\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i)\}_{i=1}^M$.

4.2 Bias Correction for Conjoint Analysis

Due to the high cost of conjoint analysis, we typically assume the size of the primary set is small. Another set of hypothetical auxiliary customers $c_{aux} = \{c_1, \dots, c_N\}$ (N should be a lot larger than M), for whom actual human choice y is not collected. We can leverage LLMs to generate synthetic choice z for c_{aux} . However, prior work [30, 14] shows that relying directly on LLM choices for demand estimation is suboptimal, since their monotonic preferences deviate from humans. It is crucial to debias LLMs to better align their preferences with those of humans. Thus, we debias LLMs by **using LLM choices z and option features x to predict counterfactual human choices $\hat{y}$, which then inform demand estimation**, rather than relying on the LLMs' choices directly.

	Features of choice tasks x	Human Choices y	LLM Choices z
Primary Customer Set	$x_{primary}$ available	$y_{primary}$ available	$z_{primary}$ available
Auxiliary Customer Set	x_{aux} available	y_{aux} Not Available	z_{aux} available

Figure 1: The general overview of the missing data situation. We assume we have task features, human choices, and LLM choices for the smaller primary set. We have task features and LLM choices for the larger auxiliary set. We don't have human choices for the auxiliary set.

The missing-data situation is shown in Figure 1. To predict the human choices of c_{aux} , we can use the primary customer data to learn the conditional distribution $\mathbb{P}(y | x, z)$. We then apply the resulting model to the auxiliary customers' (x, z) pairs to predict their likely choices $\hat{y}$. The entire auxiliary dataset $\mathcal{D}_{aux} = \{(x_i, \hat{y}_i, z_i)\}_{i=1}^N$. The combined human choices data $\mathcal{D}_{primary} \cup \mathcal{D}_{aux}$ can be used for demand estimation in common conjoint analysis settings. **The process is summarized as follows:**

- Fit a model f to approximate $\mathbb{P}(y | x, z)$, using primary dataset $\mathcal{D}_{primary}$.
- Apply the fitted model f on the auxiliary data to get a soft choice $\hat{y}_{aux}$ (distribution of choices) for auxiliary tasks.

4.3 Inference: Demand Estimation with Conjoint Analysis

One common practice for conjoint study is to estimate the part-worth $\beta \in \mathbb{R}^q$ (Appendix B.2) for each attribute by fitting a Multinomial Logit (MNL) model against choices using task features x to best approximate $\mathbb{P}(y | x)$. The best-in-class estimator of the MNL model is:

$$\beta^* \in \arg \min_{\beta \in \mathbb{R}^d} \left\{ \mathbb{E}_x [\text{KL}(\mathbb{P}(y | x) \| \sigma_y(x; \beta))] \right\} = \mathbb{E}_x \left[\sum_{j \in \mathcal{K}^+} \mathbb{P}(y = j | x) \log \frac{\mathbb{P}(y = j | x)}{\sigma_j(x; \beta)} \right] \quad (4)$$

where $\sigma_j(x; \beta) = \frac{e^{x(j)^\top \beta}}{1 + \sum_{l \in \mathcal{K}} e^{x(l)^\top \beta}}$, $\forall j \in \mathcal{K}$ is the approximation to the true choice probabilities given by the estimator.

After getting choices generated by the LLM and bias correction process, the part-worth β can be estimated from the following set of choices:

- β^* based on **all available human choices** $\{(x_{primary} \cup x_{aux}, y_{primary} \cup y_{aux})\}$. In real-world applications, we do not assume that y_{aux} is available. For evaluation purposes, we assume that we have access to it. This serves as the ground truth part-worth.
- $\beta_{primary}$ based on **primary customer choices**: $\{(x_{primary}, y_{primary})\}$
- β_{aux} based on **raw LLM choices**: $\{(x_{aux}, z_{aux})\}$
- β_{naive} based on a **naive combination of primary human choices and auxiliary LLM choices**: $\{(x_{primary} \cup x_{aux}, y_{primary} \cup z_{aux})\}$

- β^{corr} based on the **predicted human choices** of the auxiliary customers using bias correction: $\{(x_{aux}, \hat{y}_{aux})\}$

We will compare the distance between different estimators and the ground-truth estimator.

4.4 Experiment Setup and Evaluation

Datasets and Models We use both a general-purpose pre-trained LLM (Claude Sonnet 3.5v2) and an LLM (Qwen3-4B[31]) fine-tuned specifically for the conjoint task. As described in additional experiments in appendix B.3, using either a customer’s historical choices from the conjoint task or a summary of their preferences significantly improves the alignment rate between the LLM’s choices and those of human respondents. However, in a completely new conjoint task, we most likely do not have access to a person’s historical responses. Therefore, by fine-tuning the model on existing data, we hope the fine-tuned model can learn from diverse user preferences and understand a broader range of user behavior. The bias correction model is a logistic regression model. More details of the datasets and the models are in appendix B.1.

Evaluation We will measure the difference between the part-worth coefficients β estimated from different set of choices and the ground-truth part-worth coefficients β^* . One commonly used distance metrics is mean absolute percentage error (MAPE): $MAPE(\beta) = \frac{1}{q} (\sum_{i=1}^q \frac{|\beta_i - \beta_i^*|}{|\beta_i^*|})$. For the subsequent discussion, following [30], using $MAPE(\beta^{primary})$ as the benchmark, we report the change of MAPE to compare different estimators. For example, if $MAPE(\beta^{corr}) - MAPE(\beta^{primary})$ is a negative number, it suggests that the bias correction method outperforms the primary-data-only estimator. We refer this change as bias reduction (from primary estimator).

4.5 Results and Analysis

The bias reductions of different methods are shown in Table 2. The number is the lower the better. Among them, β^{tuned} and $\beta^{pre-trained}$ refer to the estimators obtained from the fine-tuned Qwen3-4B and the pretrained Claude models **after applying bias correction**. Both are β^{corr} described in Section 4.2 but LLM choices are based on different foundation models. The others are estimators derived from Claude’s raw LLM choices without bias correction, based on different choice sets as described in Section 4.2. We can see that using predicted human choices from LLM generated choices via bias correction can provide more accurate demand estimation. The bias reductions for β^{tuned} and $\beta^{pre-trained}$ are significantly more than other methods that naively using LLM choices. Directly using LLM generated choices (β^{aux}) even increases the bias, which highlights the necessity of bias correction rather than using LLM generated choices alone. A fine-tuned Qwen3-4B performs comparably to Claude (which has at least 300B parameters). We did not include results from the 4B model without fine-tuning, as the pretrained version struggles to follow task instructions effectively.

Estimators	β^{aux}	β^{naive}	β^{tuned}	$\beta^{pre-trained}$
MAPE(β)	0.11	-8.42	-16.52	-16.63

Table 2: Mean Absolute Percentage Error (MAPE) for different β estimation methods.

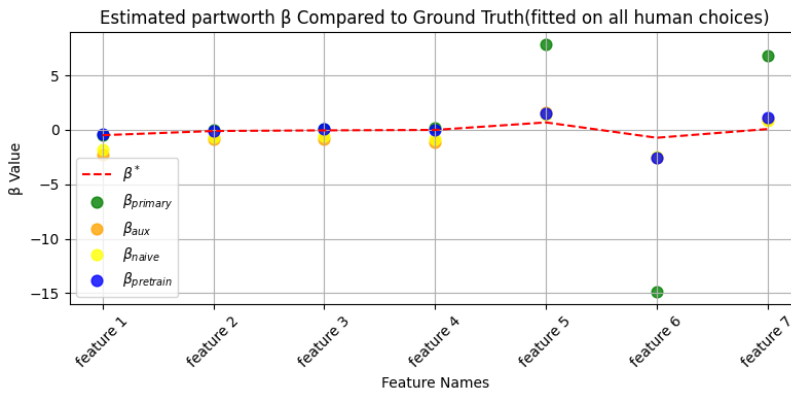


Figure 2: Partworth for different features in the conjoint task estimated based on different choice sets.

We also visualize the coefficients of the part-worth in Figure 2. We can see that the part-worth estimated using bias correction based on LLM choices from tuned Qwen3 model (blue dots) is closer to the ground-truth than other methods.

When the sample obtained from primary data differs greatly from the true customer distribution, the bias correction model can still provide an accurate estimate. These results provide preliminary evidence for the feasibility of using LLM-generated choices to augment limited human choices for accurate demand estimation.

5 Agentic Modeling for Regional Experiment

Problem description The subsequent LLM simulation is based on a regional experiment. In this experiment, new features were added to the expedited delivery options for online shopping in the treatment group. A new feature was introduced for the Same-Day delivery option, while a different new feature was added to the Next-Day and Second-Day delivery options. The results showed that the share of Same-Day delivery decreased by 60 bps, while the share of Next-Day and Second-Day delivery increased by 25 bps. The Same-Day adjustment was associated with a more consistent directional effect, whereas the modifications to the Next-Day and Second-Day options exhibited negative or mixed effects according to other metrics.

Regional experiments pose greater challenges than conjoint tasks: they must run nationwide, are costly, and can cause negative effects (e.g., revenue loss). They also require long trial periods to reach significance. To mitigate this, we propose reducing scale by either **shortening the experiment duration** or **reducing the number of regions**, and then using LLMs to simulate the remainder. In both cases, the bias correction model must perform out-of-domain prediction—over time in the first case and across regions in the second. In the following LLM simulation, we primarily focus on the effect of the treatment on the share of the Same-Day delivery. We aim to examine whether, when an LLM replaces humans in conducting the experiment, the treatment effects—measured as changes in the share of delivery options—predicted by the LLM **(1) are in the same direction as those measured in human experiments, (2) differ in magnitude (and if so, by how much), and (3) whether any such bias can be corrected.**

5.1 Generation Stage for Regional Experiment

We simulate regional demand responses by pairing region-specific personas with historically observed orders. For each ZIP3 area z , we assemble a persona set $P_z = \{p_1, \dots, p_{m_z}\}$ that captures local heterogeneity in preferences (details and examples in Appendix C.1.1). Because exact day-by-day replication of all orders is infeasible, we draw a sample of 3,806 historical orders $\mathcal{O}$ and, for each region, subsample $n_z = 42$ orders $\mathcal{O}_z \subset \mathcal{O}$. Each order is presented with a delivery menu that mirrors the original experiment: (i) an FST option (free shipping when the basket exceeds \$35; delivery time drawn from the empirical FST distribution), (ii) a standard option (strictly faster than the sampled FST time but slower than the expedited option), (iii) one expedited option—same-, next-, or second-day—selected at random subject to ZIP3 availability, and (iv) a give-up (no-purchase) option. In the treatment group, expedited delivery options were augmented with new features that differed from those in the control group, where the original configuration was retained. Acting under persona $p \in P_z$, the LLM chooses among the K options for order $o \in \mathcal{O}_z$, yielding a one-hot vector $c^l \in \{0, 1\}^K$. For a ZIP3 z , this generates triplets $\mathcal{T}_z = \{(o_i, p_i, c_i^l)\}_{i \leq n_z * m_z}$, which serve as inputs to our downstream debiasing and inference pipeline. Prompts, persona and order construction details are in Appendix C.1.

5.2 Bias Correction for Regional Experiment

Although the LLM often predicts the correct treatment direction, its effect size differs from human results, requiring bias correction similar to conjoint. Yet unlike conjoint tasks with order-level data, regional experiments lack customer-level choices, so loss must be computed at the **zip3 level**.

Let $s_z^h \in \mathbb{R}^K$ denote the vector of delivery option shares for area z observed in a small scale human experiment (by either shortening the experiment duration or reducing the experiment area). As discussed in Section 2, our mixture bias correction model then operates as follows: Given a triplet set $\mathcal{T}_z^{train}$ from a certain zip code z , it learns a weight for each triplet $t_i = (o_i, p_i, c_i^l) \in \mathcal{T}_z^{train}$. Suppose the encoded feature representation of persona p_i and order o_i is h_i . The input to the model is a batch of such representations from the same zip code $H = \{h_1, \dots, h_N\}$, then, similar to the attention mechanism [4], the weight for a pair of persona and order is

$$\theta_i = \frac{\exp(U^\top \tanh(Vh_k))}{\sum_{i=1}^N \exp(U^\top \tanh(Vh_k))} \quad (5)$$

U and V are trainable parameters. Then we compute the zip level share as $s_z^l = \sum_{i=1}^N \theta_i * c_i^l$. We then compute the KL divergence loss between the computed share and the ground-truth human share s_z^h to train these parameters.

Specifically:

$$\mathcal{L}_{KL} = \sum_z KL(s_z^h || s_z^l) = \sum_z \sum_{j=1}^K s_{zj}^h \log\left(\frac{s_{zj}^h}{s_{zj}^l}\right) \quad (6)$$

where K is the total number of available options.

In addition to the above mixture model with a weighted choice structure, for comparison, we also used a simple integrated model. Given a triplet $t_i = (o_i, p_i, c_i^l) \in \mathcal{T}_z^{train}$ as input, the model f outputs a predicted human choice $g_i = \text{softmax}(f(t_i))$ where $g_i \in \mathbb{R}^K$ for this order. Next, we aggregate all order-level predicted human choices within the same ZIP3 z to obtain the ZIP3-level share of predicted human choices $s_z^l = \frac{1}{|\mathcal{T}_z^{train}|} \sum_{t_i \in \mathcal{T}_z^{train}} g_i$. We then also compute the KL divergence to minimize the distance between the s_z^l with s_z^h . It can be seen that the integrated model does not assume a specific weighted linear structure. We implement a prompting bias correction method by injecting prior delivery option shares—estimated from the first x days of human data—into each LLM decision prompt. An example prompt is shown in C.2.4.

Time-wise and Region-wise Bias Correction As mentioned earlier, we have two ways to reduce the scale of the experiment. First, instead of conducting a full human experiment lasting more than two months, we can run a shorter human experiment for just a few days. In this case, suppose we have access to the first x days of real experimental data. We then randomly sample 1/7 of the triplets from $\mathcal{T}_z$ for each z to serve as training data $\mathcal{T}_z^{train}$, another 1/14 as validation data $\mathcal{T}_z^{val}$ with the remaining triplets used as test data $\mathcal{T}_z^{test}$ to perform inference.

The second is to conduct the human experiment in only a small subset of regions (ZIP3s), then train the model on these regions to see whether it can generalize to new regions. Suppose the set of all ZIP3 nationwide is $\mathcal{Z} = \{z_1, z_2, \dots\}$. In this setting, instead of splitting the data by time, we will divide all zip codes nationwide into two disjoint sets: $\mathcal{Z}_{ID}$ (In-Domain), and $\mathcal{Z}_{OOD}$ (out-of-domain). We aim to conduct the regional experiment only within the $\mathcal{Z}_{ID}$ areas and obtain the ground-truth human experiment labels in these regions (the share of each delivery options s_z^h). During the experiment, we assume that the human share in $\mathcal{Z}_{OOD}$ is unknown. We need to train a model to learn $P(s_z^h | \mathcal{T}_z)$ for $z \in \mathcal{Z}_{ID}$, and then apply the model to $\mathcal{Z}_{OOD}$ to obtain the predicted share of delivery options in those areas. The bias correction model here aims to learn human share based on the persona distribution and LLM choice in regions where human experiments have been conducted. We assume and hope that the learned relationship can generalize across variations in persona distributions, allowing us to extrapolate to regions where no human experiments were performed.

Full Human Experiment	W1 Human Experiment	LLM w/o BC	Prompting BC	Integrated Model	Mixture Model
−60 bps	−37 bps	−350 bps	+4 bps	−38 bps	−41 bps

Table 3: Effect on the share of Same-Day delivery across correction methods, reported in basis points. BC stands for bias correction. From left to right, we report the effect estimated from: the full human experiment; the first week of the human experiment; the LLM’s raw choices without bias correction; bias correction through prompting; bias correction through training the integrated model; and bias correction through training the mixture model.

5.3 Inference: Treatment Effect Estimation

To quantify the impact of the treatment, we employ a difference-in-differences (DiD) research design:

$$Y_{zt} = \beta_0 + \beta_1 * Treatment_z + \beta_2 * Post_t + \beta_3 * (Treatment_z \times Post_t) + \epsilon_{zt} \quad (7)$$

where Y_{zt} represents the outcome variable for ZIP3 z at time t (the share of each delivery option, each dimension of s_{zi}^l or s_{zi}^h), $Treatment_z$ is an indicator for treatment zip codes, $Post_t$ is an indicator for the post-treatment period, and ϵ_{zt} is the error term. The coefficient of interest is β_3 , which captures the treatment effect.

5.4 Experiment Setup and Evaluation

Personas and Datasets The proprietary dataset records inferred demographic details such as age band, gender, income band, zip code, and education level. We selected over 15k+ U.S.-based customers from the dataset to construct our personas. We followed the methodology outlined in [9] to generate personas based on customers’ demographic information using Sonnet3.5v2.

We assume we have the first $x = 7$ days (one week) of human data in the time-wise bias correction setting. In the region-wise setting, the sampling fraction r for $\mathcal{Z}_{ID}$ is set to 0.1. We used stratified sampling with proportional

Data source	Estimated from human experiment only in $\mathcal{Z}_{ID}$	Estimated from human experiment only in $\mathcal{Z}_{OOD}$	Train integrated model and applied to $\mathcal{Z}_{OOD}$	Train mixture model and applied to $\mathcal{Z}_{OOD}$	Estimated from national human experiments
Treatment Effect	-57 bps	-60 bps	-39 $\pm$ 9 bps	-65 $\pm$ 10 bps	-60 $\pm$ 8 bps

Table 4: Region-wise Results. Except for the national human experiment, all other experiments are the averaged results from randomly sampling zip codes and training the bias correction model 20 times. Section 5.5 includes details of the construction of confidence intervals.

allocation based on the treatment assignment in the original regional experiment and whether SSD (Sub Same-Day) delivery was launched. We randomly sample a fraction r zip3s ($0 < r < 1$) from each of the four Treatment $\times$ SSD groups to form $\mathcal{Z}_{ID}$. The remaining zip3s are assigned to $\mathcal{Z}_{OOD}$. More details of the datasets and prompts used to have the LLM make delivery choices is shown in Appendix C.2.

Bias Correction Model An input to the mixture model is a preprocessed numerical representation of a triplet (o_i, p_i) . Order features o_i include product price, delivery options related features, a product embedding generated by a Sentence Transformer [28] and other treatment related features. Persona p_i features consist of inferred demographic information such as age band, gender, income band and education level. The concatenated features of o_i and p_i are passed through a linear layer with a hidden dimension of 64, which is also the input dimension of U . The hidden dimension of the attention layer (between U and V) is 32. For a fair comparison, the integrated model also uses three linear layers, with ReLU activation between the first two layers and tanh activation between the last two layers. In addition to o_i and p_i , the integrated model’s input also includes the LLM’s choice c_i^l . The models are optimized with Adam optimizer [18]. The learning rate is $1e-5$.

Evaluation We primarily evaluate the treatment effect on the change in Same-Day delivery share in areas where SSD was launched. In the original full human experiment, the share of Same-Day delivery decreased by 60 bps after the treatment. Other outcomes in the experiment showed no significant change. Therefore, we prioritize the treatment effect on the Same-Day share in our analysis.

5.5 Confidence Interval Construction

As shown in Section 5.6 and 5.7, we report the confidence interval of the effect estimated by the bias correction model. When using LLM agents to conduct human experiments, the cost of invoking LLMs is very low compared to recruiting real human participants. In principle, this allows us to obtain an unlimited number of LLM-generated samples. Consequently, applying methods such as difference-in-differences to estimate treatment effects would, in theory, yield an infinitesimally small confidence interval. Therefore, in LLM-based experiments, **one cannot rely on sampling variance alone to measure the confidence interval**. Instead, it is necessary to **account for the variance introduced throughout the entire statistical process**, including generation, correction, and inference.

Here, we adopt a bootstrap strategy: we repeat all experiments 20 times, each time randomly sampling with replacement from the set of personas (and from regions in the region-wise experiments, which correspond to generation stage), and retraining the bias correction model (correction stage). The treatment effect is then estimated based on the bias correction model’s new outputs (inference stage). This procedure yields 20 estimated treatment effects. Assuming these estimates follow a t-distribution, we calculate their mean and construct the corresponding 95% confidence interval. This stands in contrast to the computation of confidence intervals in human experiments. As shown in the final column of Table 4, the reported confidence intervals there are based solely on sampling variance. Our bias correction method yields tighter confidence intervals in the time-wise experiments, demonstrating the robustness of our approach as well as its potential to enable early termination of experiments. In the region-wise experiments, the mixture model provides accurate effect estimates and confidence intervals that are close to those observed in human experiments, demonstrating its ability to generalize to unseen regions.

5.6 Results of Time-wise Bias Correction

As shown in Table 3, using the LLM’s raw choices without bias correction yields a treatment effect on Same-Day delivery share that is directionally consistent with the human experiment. However, this estimate differs substantially in magnitude from the actual experimental result. By training the model for bias correction using the first week of human experiment data, we can substantially reduce this gap. The mixture model produced the prediction that was closest to the human experiment. We also compute the share of delivery options predicted by each method and results (more in appendix C.3) show that the mixture model provides the most precise prediction of the shares.

How does this result imply the possibility of reducing the scale of human experiments? After applying bias correction using the first week of human experimental data, the treatment effect given by the trained model is close to the treatment effect observed in the first week of the human experiment (e.g. -41bps from the mixture model vs. -37bps from the first week human experiment). One question to ask is: why don't we simply rely on the first week of human experimental data? One major advantage of using LLMs is that the relatively low cost of simulation experiments allows for a large number of LLM-based trials, which can increase the statistical significance of the results and reduce the confidence intervals. When the human experiment is even shorter—e.g. just 1 day—resulting in wider confidence intervals or possibly statistically insignificant outcomes, can LLM augmentation combined with bias correction help produce significant experimental results? Here, we experimented with using only the **first day's** human experiment share as the ground truth for training the bias correction model. If only day one of human experimentation is conducted, the resulting treatment effect on the share of Same-Day delivery is -17 bps, with a 95% confidence interval of [-43bps, +9bps] and the p-value is 0.2049. After applying LLM-based augmentation with bias correction (without regularization) using the real share from only day one, the treatment effect is -24 bps, with a confidence interval of [-26bps, -22bps] and the p-value $< 1e^{-5}$ (Section 5.5 includes details of the construction of confidence intervals). Some remaining challenges under this setting are discussed in the appendix C.4.

5.7 Results of Region-wise Bias Correction

The main purpose of the region-wise experiment is to evaluate the model's generalization ability. In Table 4, we list the following numbers to compare: (1) the effect estimated from the **human experiment only in Z_{ID}** ; (2) the effect estimated from the **human experiment only in Z_{OOD}** ; (3) Training the **integrated model** on Z_{ID} and then applying it to Z_{OOD} to estimate the effect; (4) Training the **mixture model** on Z_{ID} and applying it to Z_{OOD} to estimate the effect and (5) the effect estimated from the nationwide human experiment (all zips).

The mixture model demonstrates stronger generalization ability than the integrated model. The estimated treatment effect by the mixture model (-65 bps) is close to that of the complete human experiment (-60 bps). The black-box integrated model produces only -39 bps. The assumed linear weighted structure may be closer to the model of the real-world setting. From the confidence intervals, we can see that the trained models both exhibit some variation, but the range of variation is within an acceptable level. In summary, the generalization ability demonstrated by the mixture model enables us to substantially **reduce the regions** required for conducting human experiments.

6 Conclusion

This paper introduces a novel framework for agentic economic modeling with LLMs. Our Generation–Correction–Inference pipeline leverages LLM-generated data while ensuring validity via systematic bias correction against real human behavior. In conjoint analysis, the bias-corrected approach yields reliable demand estimates using only a fraction of traditional survey data. The regional experiment further demonstrates scalability, replicating treatment effects from field experiments with only aggregate data. These results suggest that LLM-based economic modeling can complement traditional methods, offering substantial cost savings and enabling rapid, parallel experimentation across products and regions. Our method has already been validated in two real-world experiments as described in the paper, and additionally, we plan to test and deploy the method in our production system to further assess its effectiveness and generalizability.

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A Summary of the Two Tasks

The summary of two tasks are shown in Table 5.

	Conjoint Analysis	Regional Experiment
Decision granularity	<i>customer</i> -level choice across profiles	<i>regional</i> -level shares across the nation
Calibration leverage	customer ground-truth labels	Aggregated region-level ground-truth
Total observations	4 millions+	35k+ region $\times$ day
Synthetic scale	1.2k synthetic customer choices	17k synthetic personas

Table 5: Contrasting the two tasks and their value as a progression

B Conjoint Analysis

Conjoint experiments reveal customer preferences by asking participants to evaluate or choose among products with varying attributes, showing which features matter most and the trade-offs they are willing to make. Large tech companies have used conjoint analysis to optimize feature configuration and attribute prioritization in product launches. However, conjoint studies are costly and slow: analyzing a single product typically requires hundreds of respondents, about thousands of dollars in marginal costs, and several weeks to complete—by which time results may already be outdated. To address this, we propose running experiments on only a small subset of participants (e.g., 10% of the original scale) and using LLMs to simulate the remainder. This approach drastically reduces cost and time, but raises the key challenge of whether bias correction can yield accurate demand estimates from such limited human data.

B.1 Datasets and Foundation Models

We use the survey data from a proprietary dataset. This dataset includes massive-scale choice-based conjoint questions to collect preferences from hundreds of survey respondents over few questions relating to thousands of representative products. In each question, respondents are asked to select a store from a set of options consisting of four online stores with different delivery configurations and one physical store. There are millions of observations in this dataset. We randomly select one testing product for initial evaluation and split the customers associated with the product into a primary customer sets $c_{primary}$ and an auxiliary customer set c_{aux} as described in Section 4. We have tuned Qwen3-72B [31], where $x \in \{0.6, 1.7, 4, 14\}$ is the size of the model. For all 1500 products in the dataset, we split out 200 products for testing purpose and the other products for tuning.

B.2 MNL Model

The best-in-class estimator of the MNL model is:

$$\beta^* \in \arg \min_{\beta \in \mathbb{R}^d} \left\{ \mathbb{E}_x [\text{KL}(\mathbb{P}(y \mid x) \parallel \sigma_y(x; \beta))] = \mathbb{E}_x \left[\sum_{j \in \mathcal{K}^+} \mathbb{P}(y = j \mid x) \log \frac{\mathbb{P}(y = j \mid x)}{\sigma_j(x; \beta)} \right] \right\} \quad (8)$$

where $\sigma_j(x; \beta) = \frac{e^{x(j)^\top \beta}}{1 + \sum_{l \in \mathcal{K}} e^{x(l)^\top \beta}}, \forall j \in \mathcal{K}$ is the approximation to the true choice probabilities given by the estimator.

The estimated β coefficient of an attribute captures how much that level increases or decreases the overall systematic utility driving respondents’ choices. When different attributes are included, relative part-worths can be compared to quantify trade-offs between attributes, translating utility differences into interpretable measures of preference strength.

B.3 Prompting Experiment

We first conducted experiments to see how well the model can perform the conjoint task based on different user preferences. We attempt to incorporate customers’ profiles and historical choices into the prompt when prompting the LLM to make selections. We consider four variations:

Simple prompting the choice tasks directly The input consists of only the textual description of the choice task and the LLM will generate choices z based on that. The LLM-generated choices can be directly used to predict human choices and perform demand estimation.

Incorporating elementary customer profiles Typically, before starting the actual conjoint choice tasks, customers are asked to provide demographic information, such as age, education level, membership status, and preferred delivering methods. We incorporate this information into the prompt, aiming for the LLM to make choices based on its understanding of the behaviors of these subgroups.

Incorporating historical choices of customers For each customer c_i , we use the first n completed tasks $s_i[:n]$ as historical choices, which are included in the prompt provided to the LLM to predict the remaining tasks $s_i[n+1:]$. Each remaining task in $s_i[n+1:]$ is treated independently; that is, the LLM responds to one task at a time given the historical choices. While subgroup-level behavioral preferences may provide some signal, they often fail to capture the full extent of customer heterogeneity. A customer’s true preferences are more accurately reflected in their own historical choices. We therefore aim for the LLM to infer customer behavioral patterns from historical data when making predictions.

Using a behavioral summary derived from historical choices A potential drawback of incorporating customer historical choices is that the prompt length increases with the number of historical examples, which may negatively impact the LLM’s reasoning performance [16]. To offer a stable-length alternative, given $s_i[:n]$ and d_i , we first ask the LLM to summarize the customer behavior b_{in} in an independent run. We then incorporate b_{in} into the prompt given to the LLM.

For this part of the analysis, the number of historical choices for each customer n ranges from 1 to 8. Since n may vary, the remaining tasks for a customer to be tested also differs. To ensure a fair comparison across different numbers of historical choices, we fix the test set to be the tasks starting from the 9th task for each customer, regardless of how many historical choices are provided. For all prompting experiments, we use Sonnet-3.5v2. To assess whether the benefit arises from customers’ own historical choices, we additionally test using a fixed set of example choices for all customers in the prompt. This is equivalent to non-personalized few-shot learning. As noted above, these methods are just attempted to know how well the model can perform the conjoint task based on different user preferences. Therefore, we simply report the alignment rate between LLM-generated choices and human choices as a basic comparison instead of doing demand estimation.

The alignment rate without any profile information and historical choices is 42.8%. This serve as a baseline for prompting-related methods. Figure 3 shows the distribution of alignment rates between the LLM’s choices and each individual customer’s choices. The alignment rates vary substantially across individual customers and approximately follow a log-normal distribution. Assuming the LLM has its own default preferences and customers are rational, the figure suggests that LLM preferences differ from many of the customers and customers’ preferences are diverse.

Figure 4 shows the performance of different methods as the number of historical choices varies. A value of 0 indicates that only elementary profile information is incorporated. Overall, the alignment rate generally increases as more historical choices are added. When using the customer’s own historical choices, an average alignment rate of 52.4% is achieved with 7 examples (blue line). When using the generated behavior summary, the best alignment rate of 51.6% is obtained when 4 historical choices are used to generate the summary (orange line). In comparison, the alignment rate using only elementary profile information is only 42.4% ($n = 0$).

When using authors’ choices as the few-shot examples for all customers, increasing the number of choices improves performance. However, a performance gap persists relative to using customers’ own historical choices, suggesting that personalized preferences contribute to the observed improvements. Performance may show diminishing returns with more historical choices, either because longer prompts hinder reasoning or because additional examples offer limited new insights into customer preferences. The second explanation is more likely, as a performance plateau is also observed when using behavior summaries (green line), where the prompt length remains relatively stable.

C Regional Experiment

C.1 Regional Experiment Generation Stage Details

C.1.1 Persona Construction

As mentioned earlier, LLMs are limited by its fixed customer preferences when using their outputs for demand analysis. To simulate customer heterogeneity, we use diverse personas representing varied customer preferences. This enables

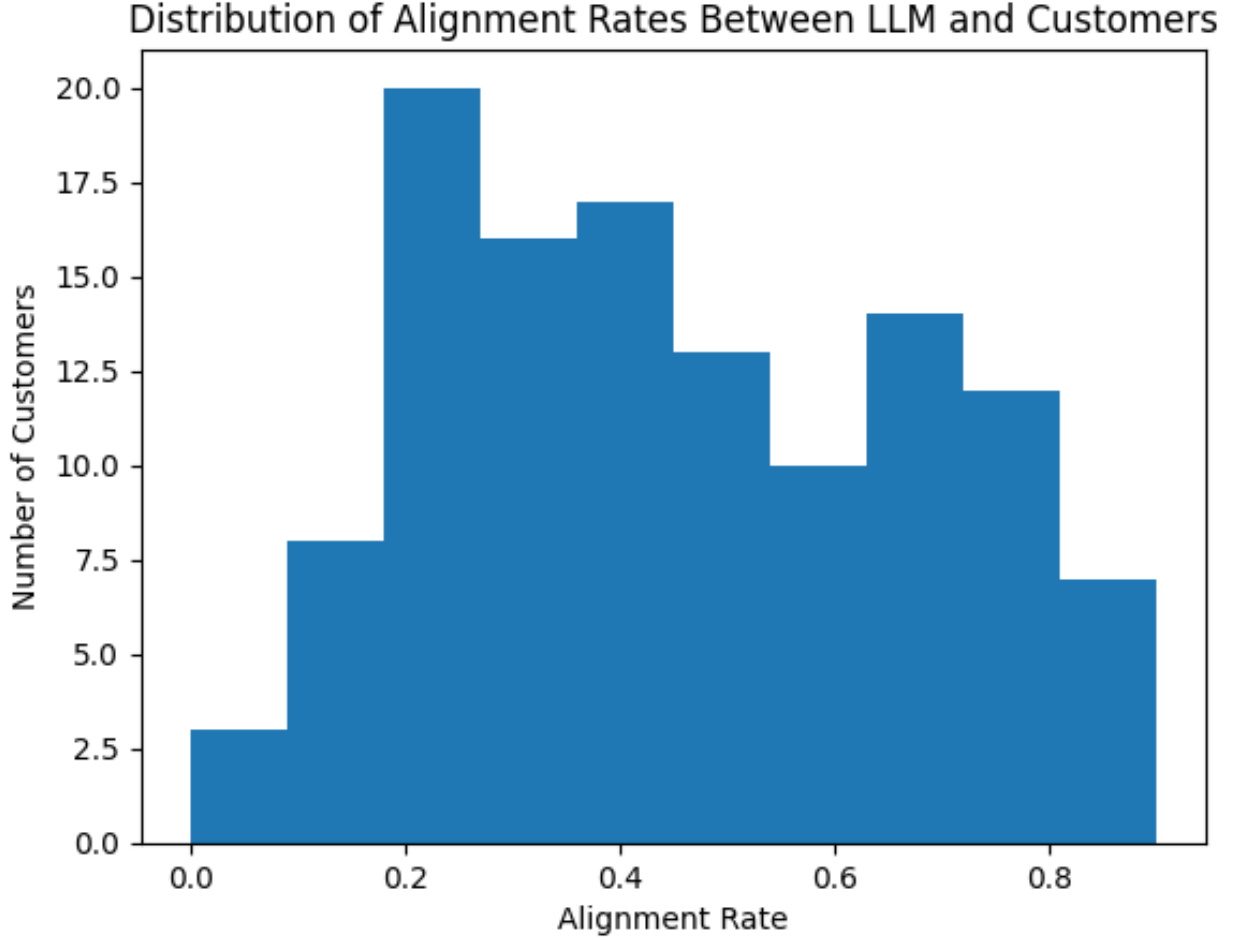


Figure 3: The distribution of alignment rates between the LLM’s choices and each customer’s choices. The alignment rates vary substantially across customers.

us to capture segment-level differences in LLM responses and improve the accuracy of experimental results. Due to differences in population distribution, the outcomes of such experiments are expected to vary significantly across regions. Ideally, for each ZIP3 area z , we aim to construct a region-specific persona set $P_z = \{p_1, p_2, \dots, p_{m_z}\}$ tailored to the demographic and behavioral characteristics of that area where m_z is the total number of available persona of area z .

Our personas are derived from the same proprietary dataset mentioned in B. The dataset records demographic details such as age, gender, income, zip code, and education level. We selected over 15k+ U.S.-based customers from the dataset to construct our personas. We followed the methodology outlined in this paper to generate personas based on customers’ demographic information using Sonnet3.5v2 (The prompt used to generate personas is provided in Appendix A.1. of the paper). An example of a generated persona is as follows:

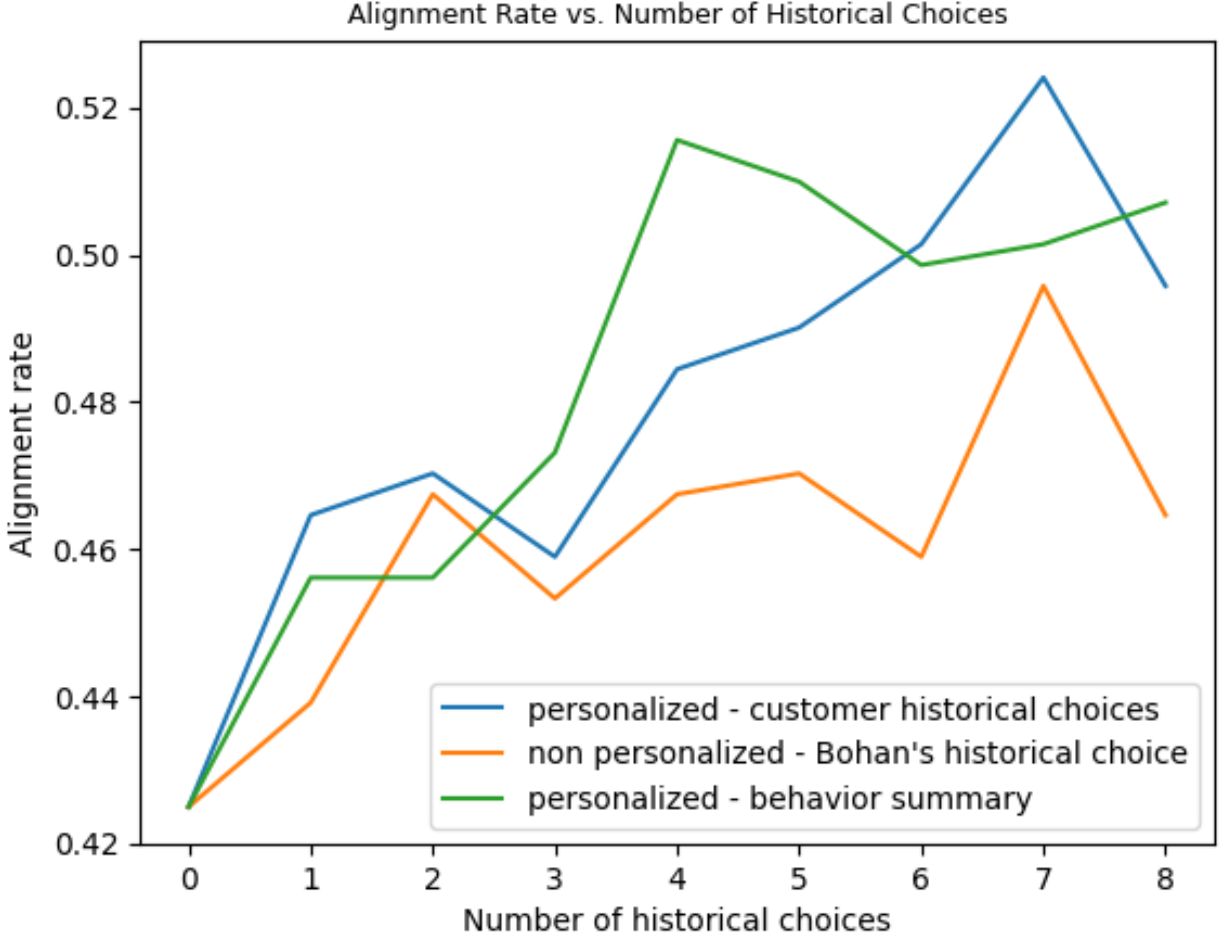


Figure 4: The performance of using customer historical choices, and customer behavior summaries, with varying numbers of historical choices. The performance is generally better with more historical choices used. Personalized methods consistently perform better than non-personalized.

[Name] is a [X]-year-old [Race] [Gender] living at [Address]. She speaks English and her educational background includes some college credits but no degree from [College]. Her date of birth is [Date]. She is currently working as a part-time retail sales associate, with an annual income of \$[Price]. She is divorced and has three adult children. She primarily uses her smartphone to stay connected with her family through Facebook and text messages, and enjoys watching cooking videos on YouTube. On her laptop, she prefers using the built-in touchpad and frequently checks her email and online banking. She is somewhat cautious about internet security and usually asks her children for help with technical issues. She spends approximately 3 hours daily on her devices, mainly in the evenings after work, and uses her phone's voice-to-text feature frequently when messaging.

The real customer information has been marked. The rest of the information was generated by the model, which is not real.

C.1.2 Order Construction

The proprietary dataset includes a table containing the actual delivery options shown to customers at checkout. However, due to the table's massive size, it is not feasible for an LLM to conduct billions of experiments. Therefore, we sampled 3,806 placed orders $\mathcal{O}$ from the historical period of the experiment. (Note that these orders were not actually placed by

the personas described above.) To simplify the experiment, we excluded all orders containing multiple products from the sample.

These orders also come from ZIP3s across the country, with an average of 4.4 orders per ZIP3. One potential concern may arise here: Why not sample more orders? One reason is that the original proprietary dataset is quite messy—many orders in the proprietary dataset cannot be matched to actual placed orders in other tables, which contain actual product information. Moreover, if we hope to apply this framework to other real-world experiments, we often do not expect having access to such detailed, fine-grained data. Given that our final analysis is conducted at the ZIP3 level and involves aggregating over orders to compute averages, sampling a manageable subset of orders is a practical approach.

The sampling of the set $\mathcal{O}_z$ ensures inclusion of all sampled orders that were actually placed in the given ZIP3. However, as mentioned above, since each ZIP3 has only about 4.4 orders on average, the remaining of the 42 orders inevitably come from other regions. The number $n_z = 42$ is chosen to simulate a 6-week LLM experiment and also manageable for bedrock quotas. Basically, we assume that one LLM experiment day corresponds to having all personas in a given ZIP3 place the same order and make a delivery option choice (In other words, although n_z represents the number of orders, it also effectively corresponds to the number of simulated LLM experiment days). This number is can be adjusted as needed. For example, we can vary this number to examine how many LLM-simulated experiments are needed to produce reliable and statistically significant treatment effect estimates.

The delivery options for each order were configured as follows: If the order value exceeded a specified threshold (e.g., \$35), a free-shipping FST option was included, with a delivery day sampled from the empirical FST distribution. A standard delivery option was then added, with a delivery time constrained to be shorter than the sampled FST time. An expedited option—same-day, next-day, or second-day delivery—was also included, selected at random subject to ZIP3 availability (if same-day service was unavailable, only next- or second-day options were used). In the treatment group, the expedited delivery options were configured with new features that differed from those in the control group, where the original configuration was retained. Finally, a give-up option was included, allowing the customer to choose not to place the order.

C.1.3 Persona-Order Interaction and LLM Choice Generation

The basic idea to using an LLM to conduct this randomized experiment is to prompt the LLM, acting as a given persona, to choose which delivery option they would select when placing a specific order.

For each ZIP3 z , we sampled n_z orders $\mathcal{O}_z = \{o_1, o_2, \dots, o_{n_z}\}$ from the pool $\mathcal{O}$. Next, we have all personas in P_z select a delivery option for each of the n_z sampled orders in their corresponding ZIP3. Let c_{ij}^l be the LLM choice when acting as $p_j \in P_z$ to place order $o_i \in \mathcal{O}_z$. Here, a choice $c \in \mathbb{R}^K$, where K is the total number of available delivery options. A value of $c_k = 1$ and $c_q = 0$ for $q \neq k$ indicates that the LLM selected the k^{th} option. For simplicity, the subscripts indicating the dependence of o , p and c on z are omitted. A simplified process describe above is as follows:

```

for  $o \in \mathcal{O}_z$  do
  for  $p \in P_z$  do
    LLM acts as  $p$  to select one delivery option  $c$  when placing order  $o$ .
  end for
end for

```

This process generates a list of triplets

$$\mathcal{T}_z = [(o_1, p_1, c_{11}^l), \dots, (o_1, p_{m_z}, c_{1m_z}^l), \dots, (o_{n_z}, p_1, c_{n_z1}^l), \dots,$$

$(o_{n_z}, p_{m_z}, c_{n_zm_z}^l)]$ for each z . For simplicity, let $t_i = (o_i, p_i, c_i^l)$ be the i^{th} triplet in $\mathcal{T}_z$ for $1 \leq i \leq n_z * m_z$. The prompt used to have the LLM make delivery choices is shown in Appendix C.2.2, which is similar to the one used in the earlier conjoint experiment. In the experiments, n_z is set to 42 for all zip3 area.

C.2 Regional Experiment Setup

C.2.1 Order Datasets

Another proprietary dataset contains the real delivery options shown to customers at checkout. We sample 3806 real orders as $\mathcal{O}$ from the table.

C.2.2 Prompt given to LLM to make delivery option choice

The confidential information has been replaced.

Suppose you want to buy a [Product]. The price is \$[Price]. We will ask you one question about buying the tire mount with different delivery options. Consider the product itself, delivery options when making your choice. You should act like this person when purchasing tire mount:

[Name] is a [X]-year-old [Race] [Gender] living at [Address]. She speaks English and Spanish, and her educational background includes completing one year at [College], where she is pursuing a degree in [Major]. Her date of birth is [Date]. She is currently working as a junior software developer at a tech startup, with an annual income of \$[Salary]. She is single and has no children. She is highly tech-savvy, spending approximately 6–8 hours daily on her MacBook Pro for both work and personal use. She prefers using dark mode on all her devices and frequently uses productivity apps like Notion and Trello. On her iPhone 14, she actively engages with social media platforms, particularly Instagram and TikTok, where she follows tech influencers and programming tutorials. When browsing the internet, she uses Chrome as her primary browser with multiple extensions for web development, and frequently participates in coding forums and online programming communities. She typically keeps multiple tabs open while working and uses cloud storage services for file management.

Imagine you had the following delivery options to buy the tire mount. Which one would you prefer? You can assume the shopping options are identical in all respects not listed here.

A (Delivery Option 1): [Details are hidden]

B (Delivery Option 2): [Details are hidden]

C (Option 3): Prefer not to buy it.

Which option would you choose as this person? Your response should be a single letter A, B, or C.

The real customer information has been marked. The rest of the information was generated by the model, which is not real. As shown, each triplet—consisting of an order, a persona, and a delivery option—is specified within the prompt. In the current experiment, there are 638484 such prompts in total.

C.2.3 Region-wise Prediction ZIP Split

To sample Z_{ID} , in the original regional experiment, there were 229 treatment zip3 areas with SSD launched and 220 without SSD. In the control group, there were 245 zip3 areas with SSD and 212 without. We randomly sample a fraction r zip3s ($0 < r < 1$) from each of the four Treatment $\times$ SSD groups to form Z_{ID} . The remaining zip3s are assigned to Z_{OOD} . Here, we implicitly assume that SSD is a source of region-wise heterogeneity. This aligns with the original regional experimental design, where the treatment effect on the delivery options differed depending on whether SSD was launched.

C.2.4 Prompting Bias Correction

Here is an example prompt for bias correction:

We calculated the prior distribution of delivery preferences based on historical data of customers placing orders within the same zip code.

This data includes all product categories and all orders, excluding cases where customers decided not to complete the purchase (“Prefer not to buy it”).

From this filtered dataset (i.e., only including orders that were actually completed), we computed the following distribution of delivery preferences:

- Same-Day delivery: X%
- Next-Day or second-day delivery: Y%
- Longer delivery time (e.g., 3+ days): Z%

These percentages sum to 100% and represent customer preferences conditional on purchasing (i.e., excluding those who chose not to buy).

We do not observe or estimate the share of customers who chose not to buy the item (the last option), so this “give-up” rate is unknown.

Please take this prior into account when making a decision about the delivery option, assuming the customer has some typical delivery preference behavior based on local historical data.

This text was added before the choice question in the prompt shown in C.2.2.

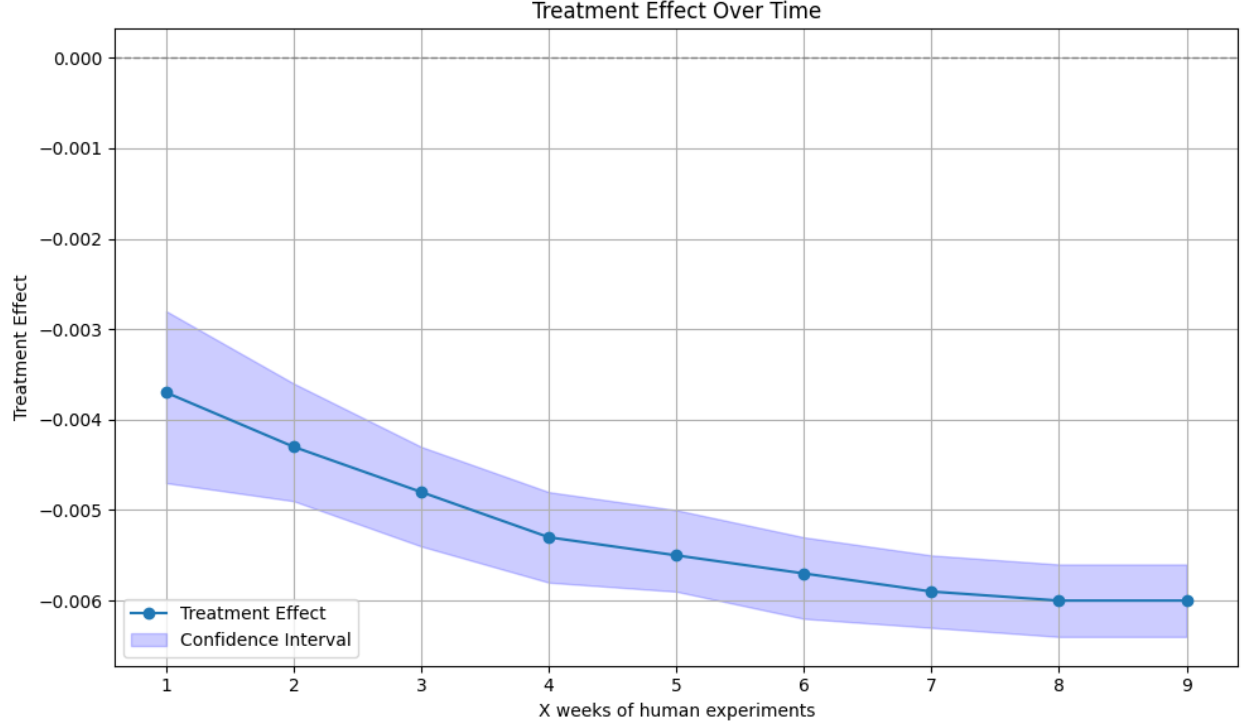


Figure 5: The treatment effect on the share of Same-Day delivery, estimated using different numbers of weeks of human experiment data

C.3 Additional Results

C.3.1 Evaluation of share prediction

We compare four groups of delivery option shares against corresponding ground-truth share from human experiments: 1) Shares in ZIP3 areas where SSD was launched and assigned to the treatment group 2) Shares in ZIP3 areas where SSD was launched and assigned to the control group 3) Shares in ZIP3 areas where SSD was not launched and assigned to the treatment group 4) Shares in ZIP3 areas where SSD was not launched and assigned to the control group. We calculate the MAPE between the ground truth share and estimated share of different groups.

C.3.2 Results

As shown in Table 6, using a trained model for bias correction significantly improves the accuracy of share prediction, while using prompts for bias correction increased the bias. This is likely because one option accounts for 97% of the choices, and prompting encourages the model to overwhelmingly favor this “safe” option. The mixture model achieved the best predictions in three groups.

	Before Bias Correction	Prompting Bias Correction	Integrated Model	Mixture Model
SSD, T	0.2518	0.6794	0.1109	0.0518
SSD, C	0.6519	0.6817	0.1414	0.1254
Non SSD, T	0.3021	0.5094	0.0928	0.1777
Non SSD, C	0.1366	0.5102	0.0939	0.0768

Table 6: Performance of share prediction under different bias correction strategies for SSD and Non-SSD (Treatment and Control groups). We use the first week’s human experiment as the ground truth. The lower the number, the better the performance.

C.4 Further Analysis of Time-wise Prediction

Even though the increased number of simulations enhances the significance of the results, there remains a gap between the treatment effect estimated using bias correction based on a short-period human experiment and that derived from the full human experiment. This reflects a limitation of current time-wise experiments: they are unable to model the temporal changes of the regional experiment. Such regional experiments are typically subject to temporal effects. For example, Figure 5 shows the treatment effect on the share of Same-Day delivery estimated using different numbers of weeks of human experiment data.

As time progresses, the treatment effect tends to stabilize (though future changes cannot be ruled out), and the confidence interval narrows. The share of Same-Day delivery also fluctuates over time throughout the experiment (its change reached up to 40% of the initial value). Since our LLM experiment does not incorporate temporal modeling, it cannot capture these temporal dynamics. Achieving temporal modeling in LLM experiment requires OOD prediction, since during LLM experimentation and bias correction, we cannot assume knowledge of such temporal trends. To make forward-looking predictions, we need to consider whether the changes in these variables are driven by people adapting to the effects of the experiment itself, or by external factors such as new users entering the system, holidays, or other contextual shifts? This is inherently a challenging task.